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ISOLATION, PROPAGATION AND CULTIVATION OF GREY OYSTER MUSHROOM (*Pleurotus ostreatus*)

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Abstract

Grey oyster mushroom (*Pleurotus ostreatus*) is one of the favorite edible mushrooms commonly used for food processing and medicinal purposes. In the cultivation process of oyster mushroom, yield and biological efficiency are highly dependent on quality of strain, propagation methods, cultivating substrates and methods. In this study, the isolation, selection of strains, spawn propagation and experimental cultivation of grey oyster mushroom have been conducted. From three sources of fruiting bodies, three strains of *P. ostreatus* were isolated. In which, strain OM3 showed the strongest growing ability on PGA medium as well as PG liquid medium: colony diameter reaching 8.8 cm on PGA (propagation level 1); 860.3 spherical colonies /100 mL and 43.35 g dry biomass/L (propagation level 2). The liquid spawn of *P. ostreatus* OM3 was selected for grain spawn production and experimental cultivation forming fruiting bodies. The propagation medium consisting of 99% paddy + 1% $CaCO_3$ was the most suitable for producing grain spawn (propagation level 3): the mycelium covered all paddy grain substrate within 7 days. The component of substrates including 48% sawdust + 48% rice straw + 3% rice bran + 1% $CaCO_3$ was the best for growth and formation fruiting bodies of *P. ostreatus* OM3: the time to cover the substrate by mycelium and the time to form fruiting bodies were very short (14.07 days and 23.37 days), high mushroom yield (reaching 916.8 g/ 1.5 kg dry substrates), high biological efficiency (reaching 61.12%). This is a complete study about grey oyster mushroom including selection of strain, propagation level 1, 2 and 3 on different mediums and on different cultivating substrates for obtaining fruiting bodies with high yield and biological efficiency.

Keywords: *Pleurotus ostreatus,* oyster mushroom, propagation, cultivation, fruit body.

1. INTRODUCTION

Oyster mushroom (*Pleurotus ostreatus*) is belong to class Basidiomycetes, one of the popular edible mushrooms around the world. The fruiting bodies of oyster mushroom are soft, and express colour variation depending on their age and environmental conditions. The stem of fruiting body is being off centre and its cap is similar in shape to an oyster shell. Oyster mushrooms can be found in colors including blue, yellow, pink, brown and gray. Gray oyster mushroom refers a type of oyster mushroom with a gray cap.

P. ostreatus is one of the most widely cultivated edible mushrooms due to its economic (food source), ecological (waste remediation agents), and medicinal value (biocompounds supplying source) (Da Luz et al., 2012). Oyster mushroom is a popular and well-loved food source because they contain high protein, fiber, vitamins and minerals contents; low fat, salt, calories and cholesterol contents (Caglarirmak, 2007). In addition, they also contain bioactive compounds with potential in pharmaceuticals and medicine such as anti-cancer, anti-cholesterol, anti-tumor, immune system support, antibacteria and antioxidant compounds,... (Chorvathova et al., 1993; Sivrikaya Hüseyin et al., 2002; R. Mary Josephine, 2015; Mihai et al., 2022).

P. ostreatus can grow at wide temperature range and adapt to different climatic conditions. All of the agricultural and forestry residues that are rich in cellulose such as rice straw, wheat straw, cotton waste, sawdust, coconut waste, wood stalk and additives (inorganic and organic fertilizers) are main materials for growth of oyster mushroom (Shah et al., 2004; Piškur et al., 2011; Girmay et al., 2016). The advantage of cultivation *P. ostreatus* is that the growth time is shorter than other edible mushrooms, it can grow on sterilized or unsterilized substrates. In addition, it brings high profits (high substrate conversion rate), and is less susceptible to attack by diseases and pests (Sánchez, 2010).

In Vietnam, oyster mushrooms can be grown all year round depending on the temperature conditions of each region and each variety group, but the most favorable season is from August to April of the following year. The substrates for cultivation oyster mushrooms is mainly straw, cotton waste, sawdust,... which have been sterilized or fermented (compost) (Truong, 2004; Truong Binh Nguyen et al., 2019). In the process of cultivating oyster mushroom, improving biological efficiency, shortening cultivation time, isolating good quality strains and chosing propagation methods are of great important. Therein, the method of propagation by liquid fermentation and using liquid spawn had advantages in shortening mushroom cultivating and spawn running time, reducing the rate of contamination and the time to form fruiting compared to the traditionally propagation (Vladimir, 2012; Pham et al., 2018; Pham Thi Thu, 2018). This paper presents the results of research on isolating, propagating spawn and experimental cultivating *P. ostreatus* to develop quality varieties while improving biological effectivity.

2. MATERIALS AND METHODS

2.1. Material

Fruiting bodies of grey oyster mushroom: Purchased from market (S1), supermarket (S2) and mushroom growers (S3) in Hanoi, Vietnam. S1 was not determined date of harvesting, S2 was 1 day after package and S3 was just harvested.

Culture medium: Solid PGA medium (potato extract 200 g/L, glucose 10 g/L, agar 20 g/L, distilled water 1000 mL) and liquid PG medium (potato extract 200 g/L, glucose 10 g/L, distilled water 1000 mL). Mediums were sterilized for 20 minutes, 121 ^{o}C at 1atm.

Spawn medium: Paddy grains were soaked overnight, boiled until shell cracks and sterilized in 90 minutes at 121 ^{o}C, 1 atm. The addition of rice bran depended on each treatment.

Sawdust, rice straw: Sawdust and rice straw were supplied from mushroom farm in Vu Thu district, Thai Binh province, Vietnam.

2.2. Methods

Isolation of oyster mushroom strains from fruiting body: using sterilized papers soaked in 70% alcohol to wipe the fruiting body; cutting the fruiting bodies in half lengthwise; cutting small pieces (0.5x0.5 cm in size) at the junction between the cap and the stem; placing the pieces on the petri plate containing sterilized PGA medium; incubating in 25 ºC to monitor the growth of mycelium; chosing samples with homogenous mycelium, without contamination (Le Duy Thang & Tran Van Minh, 2005).

Propagation level 1 in PGA solid medium: Cutting a piece of agar containing mycelium (0.5x0.5 cm in size) and transfering to sterilized PGA medium plate; incubating them in 25 ºC; monitoring the growth of mycelium for 7 days by measuring colony diameter. The experiment was repeated 3 times.

Propagation level 2 in PG liquid medium (liquid spawn): Cutting 2 pieces of agar containing mycelia of the culture level 1 (0.5x0.5 cm in size) and transfering into a conical flask containing 100 mL of PG liquid medium; shaking at 150 rpm and 25 °C; monitoring the growth of mycelium through the number of spherical colonies formed in 5 days of culture. Then, the cultured solution was centrifuged to collect biomass. The biomass was washed with distilled water and dried at 70 ºC to constant weight (g/L). The experiment was repeated 3 times.

Propagation level 3 in paddy grain medium (for graining spawn): Using three fomulars including P3.1 (89% paddy + 10% rice bran + 1% $CaCO_3$); P3.2 (94% paddy + 5% rice bran + 1% $CaCO_3$); P3.3 (96% paddy + 3% rice bran + 1% $CaCO_3$) and P3.4 (99% paddy + 1% $CaCO_3$). Adding 5% liquid spawn into 100 g of each paddy grain medium (thickness of the cultural medium was 10 cm). Monitoring propagation of the mycelium.

Cultivation of oyster mushroom: Using substrates in experimental fomulars as following: FB1 (60% sawdust + 40% rice straw) and FB2 (48% sawdust + 48% rice straw + 3% rice bran + 1% $CaCO_3$). The sawdust and rice straw were mixed well with lime water (4 kg lime powder + 1000 L water) at ratio 100 kg materials + 60 L lime water (humidity reaches 60-70%). Covering a tarp on the substrate pile for composting within 7 days. The composted material was mixed with rice bran and $CaCO_3$ in correct ratio and packed into plastic bags (2.5kg materials/bag that was equipvalent to 1.5 kg of dry substrate /bag); 30 bags was used for each formula. The material bags were put in the hot steam oven overnight. Taking out and waiting for the material to cool down, sowing grain spawn on the outer edge of the material layer in the plastic bag for each layer of compacted material 5 cm thick until 2.5 kg of material and 30 g of grain spawn/bag (spawn inoculation rate 2% by dry meterial weight) were completed. Monitoring the time it takes the mycelium to cover the entire substrate and determining biological efficiency. Biological efficiency (%) = FWm/DWs×100%

In which: FWm - The fresh weight (g) of the harvested mushrooms; DWs - The dry weight of the substrate (g) (Girmay, 2016).

Statistical Analysis: Data were analyzed by Excel 2010 and Irristat 5.0 software.

3. RESULT AND DISCUSSION

3.1. Isolation of oyster mushroom strains from fruiting bodies

In this experiment, the cutting samples from the fruiting bodies were cultured on PGA medium for isolating strains. The result of isolation was showed in Table 1 and Figure 1.

Table 1. Colony diameter and mycelial running speed of cultures isolated from the fruiting body

Samples	Isolates	Colony diameter (cm)	Speed of mycelial running (cm/day)
S1	OM1	2.4±0.15^{b}	0.60±0.17^{b}
S2	OM2	3.3±0.45^{a}	1.03±0.12^{a}
S3	OM3	3.9±0.38^{a}	1.13±0.05^{a}
5% LSD		0.76	0.39

Note: Mean values followed by the different letter within a column are statistically different for 0.05 significant level.

Mushroom fruiting bodies are formed from mycelium. After the growth phase of the mycelium, the fruiting body formation phase will take place in appropriating conditions. The formation of the fruiting body is started by the tight entwining of the mycelium to form the germs of fruiting body and mushroom eggs. The mushroom eggs are differentiated into the main parts of the fruiting body such as stem, cap, and lamina... Oyster mushrooms (*Pleurotus*) have basidia containing basidiospores that are produced on the surface of the laminae. The basidium is made up of the top of the mycelium. The basidiospores form a layer on the surface of the lamina, so the spores also form a coating on the surface of the lamina. Spores are sexual, they germinate into primary mycelium in appropriate conditions. The primary mycelium forms secondary hyphae and develops into a mycelium network spreading throughout substrate or the culture medium (Nguyen Thi Sau,

2009). Oyster mushroom can be isolated from any parts of fruiting bodies. However, it is best to isolate oyster mushroom strains from the inner part that is located at the intersection between cap and stem of the fruiting body.

According to the result of table 1, the mycelium from different samples exhibited different growth: the mycelium from S3 sample (isolate OM3) grew best after 5 days with the colony diameter reached 3.9±0.38 cm, followed by S2 sample (isolate OM2) with the colony diameter of 3.3±0.45 and S1 sample (isolate OM1) with the lowest colony diameter of 2.4±0.15 cm. These isolates were stored in slant PGA medium for next experiments.

Because the colony diameter of the mycelium from S3 sample was the largest, the propagation of mycelium of isolate OM3 was also the highest (1.13±0.05 cm/day), followed by isolate OM2 (1.03±0.12 cm/day) and the lowest was the speed of the mycelial running of isolate OM1 (0.60±0.17 cm/day).

The colony diameters of isolates from S2 and S3 samples were not different, but they were significantly different from isolate from S1 sample.

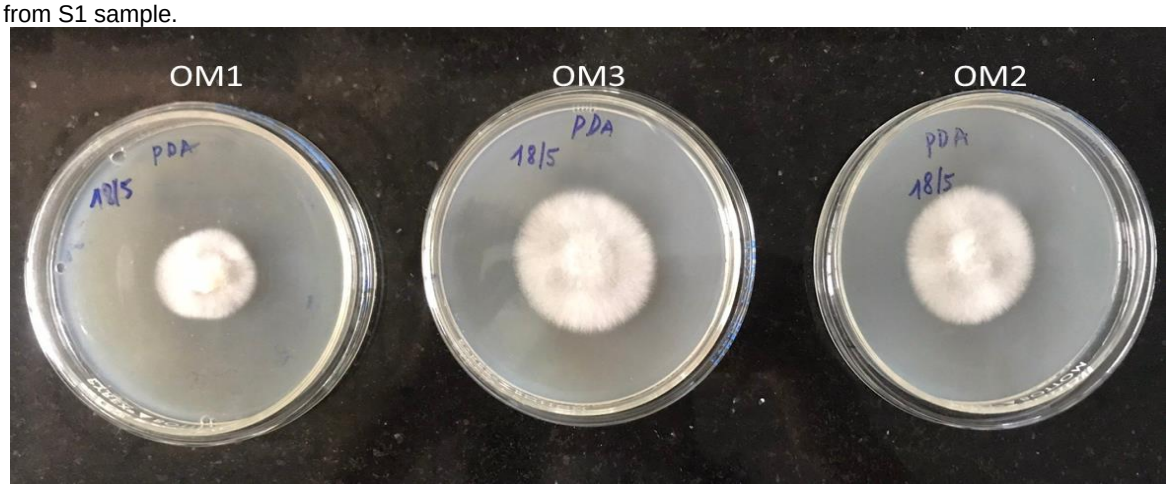


Figure 1. The mycelium of oyster mushroom isolated from the fruit body
OM1: The mycelium isolated S1 sample;OM2: The mycelium isolated from S2
OM3: The mycelium isolated from S3

3.2. The propagation level 1 in PGA solid medium

To select strains with strong growth ability**,** the slant cultures were transferred to petri plates of PGA medium for propagation level 1. Two days after culturing, the mycelium recovered from the injury and began to spread on the surface of the agar plate. It was observed that the spreading of the mycelium was very fast and had radioactive pattern. The growth of mycelium was determined by colony diameter within 7 days. In fact, after 7 days, the mycelium of isolate OM3 had covered full the surface of the agar plate (Figure 2).

Table 2. Colony diameter of the isolates after 7 days of culture

Isolates	Colony diameter (cm)	Speed of mycelial running (cm/day)
OM1	7.3±0.17^{b}	1.38±0.08^{b}
OM2	7.6±0.30^{b}	1.53±0.06^{ab}
OM3	8.8±0.20^{a}	1.63±0.04^{a}
LSD 5%	0.59	0.17

Note: Mean values followed by the different letter within a column are statistically different for 0.05 significant level.

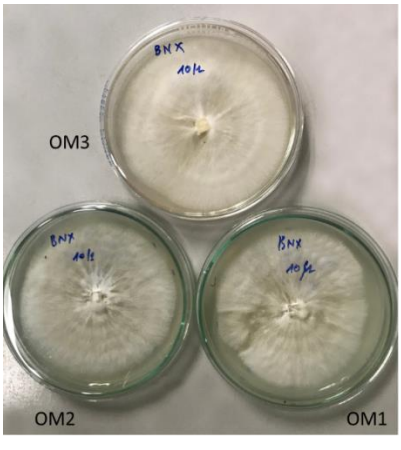


Figure 2. The mycelium of isolates OM1, OM2 and OM3

Table 2 also showed that the growth of isolate OM3 in propagating level 1 was faster and stronger than the other isolates. Colony diameter and the speed of mycelium propagation of OM3 were the highest, reached 8.8±0.20 cm and 1.63±0.04 cm/day, respectively. The colony diameter of strain OM3 had significant difference to colony diameters of isolate OM1 and OM2. The growth rate of isolate OM3 was slightly different from isolate OM2 and significantly different from isolate OM1.

The fruiting body sample (S3) was purchased from household at the right time of harvest, while fruiting body samples S1 and S2 purchased from market and supermarket had been harvested previously. This may be the reason why strain OM3 grew better than strains OM1 and OM2. This also suggests that, to isolate oyster mushroom strains from fruiting bodies, it is necessary to isolate as soon as possible after harvest.

3.3. The propagation level 2 in PG liquid medium

The mycelia of three strains of *P. ostreatus* were used for propagating level 2 in PG liquid medium. In the liquid medium, the mycelia of *P. ostreatus* were entwined together to form beads with spherical shape (spherical colonies) (Nguyen Thi Bich Thuy, 2014; Pham Thi Thu et al., 2018). The spherical colonies were distributed uniformly in the liquid medium, without stratification, and they tended to settle down when the shaking was stopped.

Table 3. The growth of *P. ostreatus* OM1, OM2 and OM3 in liquid medium

Isolates	Number of the spherical colonies (/100 mL)	Dried biomass (g/L)
OM1	803.7±8.02^{b}	39.99±0.206^{b}
OM2	812.3±15.50^{b}	40.51±0.80^{b}
OM3	860.3±9.87^{a}	43.35±0.88^{a}
LSD 5%	30.32	1.64

Note: Mean values followed by the different letter within a column are statistically different for 0.05 significant level.

The colony density of strains OM1, OM2 and OM3 grown in liquid medium reached 803.7, 812.3 and 860.3 colonies/100 mL, respectively. Among them, the colony number of strain OM3 was significantly higher than the remaining strains (p <0.05). Similar to the number of colonies, the dried biomass of strain OM3 was the highest (43.35 g/L), followed by OM2 (40.51 g/L) and OM1 (39.99 g/L). The difference in the number of colonies as well as dried biomass between OM1 and OM2 was insignificant.

The colony number of *P. ostreatus* OM3 in our study is almost equivalent to the number of colonies of *P. citrinopileatus* golden oyster mushrooms in the research of Pham Thi Thu et al. (2018) (> 90 spherical colonies/10 mL, i.e. > 900 colonies/100 mL when propagation in liquid medium containing fructose).

The spherical colonies of *P. ostreatus* had a milky white color, the colony surface was not smooth, the size was very small and relatively similar in all three strains.

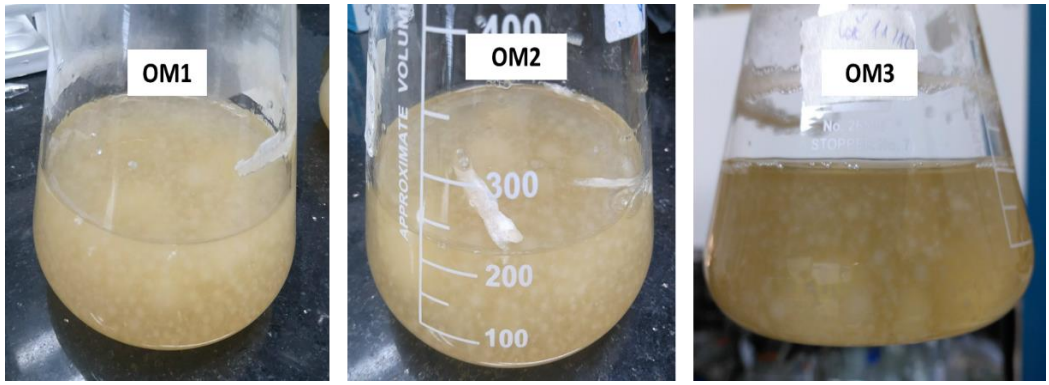


Figure 3. The spherical colonies of the strains *P. ostreatus*

3.4. Propagation level 3 in paddy grain medium

To select ingredients and appropriate ingredient ratios for propagation level 3 (grain spawn) of *P. ostreatus*, four fomulars with different ingredients were used. All of these formulas contained mainly paddy grain (from 89 to 99%).

Cultivation of edible mushrooms using liquid propagation technology has been shown to have many advantages over traditional propagation technology, such as shortening the cultivation time, reducing infection rate and better cultivation vigor. Culturing in liquid medium also has a positive effect on the growth of fungal mycelium and the biosynthesis active compounds (Abdullah et al., 2013). Morever, in a previous research on milky mushroom, the growth of mycelium using liquid culture source was stronger than solid culture source, the spawn running time reduced 2 days. This has shortened the time to propagate and cultivate mushroom (Ha Thi Quyen et al., 2022). Thus, the liquid spawn was used in this experiment and strain OM3 was chosen because of its best growing ability.

Among the four fomulars, P3.4 (99% paddy grain + 1% $CaCO_3$) had the strongest mycelium spread. After 7 days of the culture, the mycelium running of P3.4 coverd full paddy seeds (9.73 cm in the thickness of medium), the next was P3.3 (96% paddy grain + 3% rice bran + 1% $CaCO_3$) with 9.10 cm, P3.2 (94% paddy grain + 5% rice bran + 1% $CaCO_3$) and P3.1

(89% paddy grain + 10% rice bran + 1% $CaCO_3$) were 7.53 cm and 7.1 cm, respectively. In days 3 and 5, the mycelium of P3.4 formula also grew faster than the other formulas. This proved that medium including 99% paddy grain + 1% $CaCO_3$ was suitable for *P. ostreatus* mycelium. It is possible that in the formulas P3.1 and P3.2, the bran ratio was quite high (5% and 10%) so it could not create the porosity in the medium, while the mycelium needs aeration condition to grow. And the rice bran ratio of P3.3 was lower than P3.2 and P3.1 (3%) so the mycelium grew better.

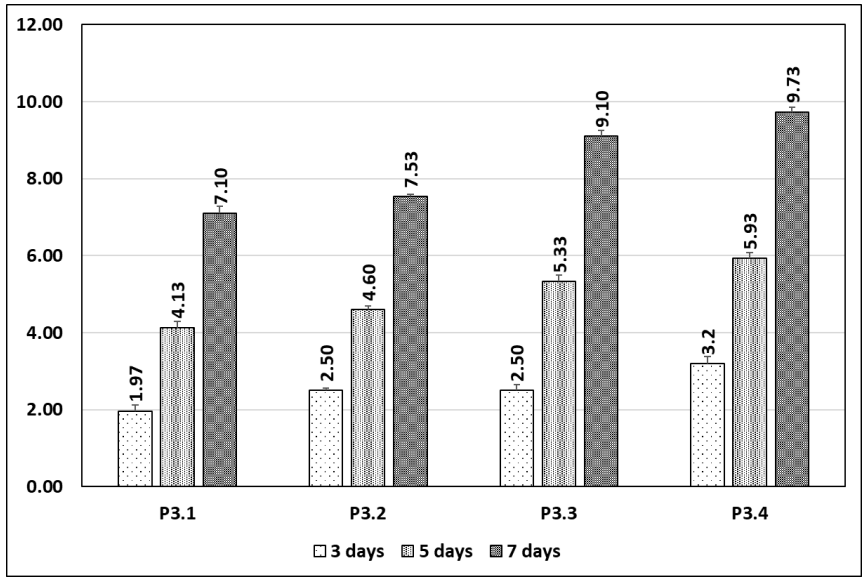


Figure 4. The growth of liquid spawn of strain OM3 in different formulas

The error bars indicated standard error of three replicates. The difference was statistically significant between formulas determined with p-value < 0.05.

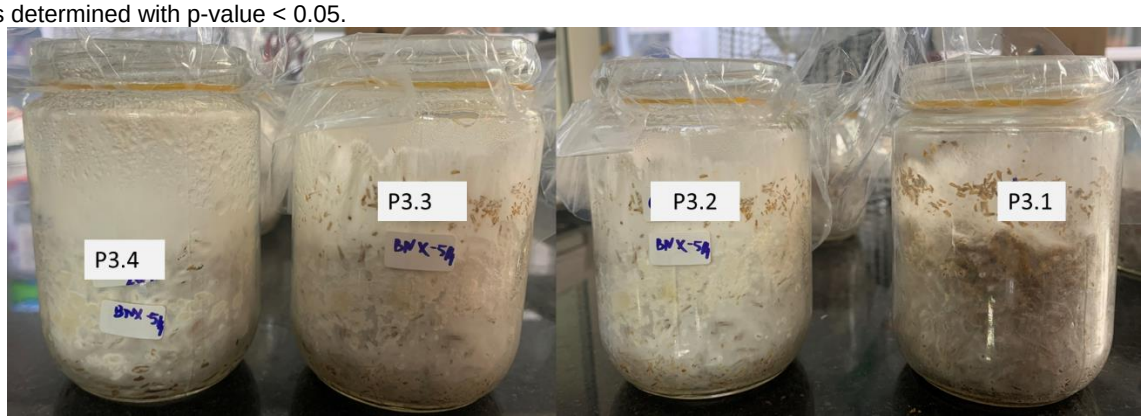


Figure 5. The mycelium level 3 on the paddy grain mediums

The research to find ingredients and ratios of raw materials to produce highly effective mushroom spawn has been of interest. There have been many suitable substrates for developing mycelium, of which the main ingredients are rice, wheat, corn kernel, and millet. These ingredients are used individually with different ratios or supplemented with a few of nutritional sources such as corn bran, rice bran, $CaCO_3$ (Tran Thanh Thy & Le Van Vang, 2022; Nguyen Thi Bich Thuy et al., 2019). The research of Nguyen Thi Bich Thuy et al. (2019) showed that medium including 99% paddy grain + 1% $CaCO_3$ was the most suitable to prepare spawn of *P. eryngii*, similar to this study for *P. ostreatus*.

3.5. The cultivation of *P. ostreatus* OM3

The number of days for mycelium to fill the substrate in FB2 formula (48% sawdust + 48% rice straw + 3% rice bran + 1% $CaCO_3$) was less than in FB1 formula (60% sawdust + 40% rice straw) about 7 days. In this study, the mycelia covered all the substrates in the bags in the range from 14.07 days (fomula FB2) to 21.10 days (formula FB1). The number of days to form fruiting bodies of FB2 (23.37 days) was also smaller than FB1 (31.97 days). The average mushroom yields per bag obtained from FB1 and FB2 formulas were 807.7 and 916.8 g/bag, respectively. Biological efficiency was calculated based on mushroom yield, ranging from 53.85% (FB1) to 61.12% (FB2).

The time for covering entire substrates by mycelium, time for forming fruiting bodies, the yield and biological performance in the cultivation of oyster mushrooms in particular and edible mushrooms in general depend highly on the cultivating substrate and the vigor of the strain. In this study, the strain OM3 showed a good vigor and was quite suitable to grow on the substrates consisting of sawdust (48%), rice straw (48%) and supplemented with rice bran (3%) and $CaCO_3$ powder (1%).

There are many different substrates used for cultivating edible mushrooms. They can be used lonely or mixed together in certain proportions. Each type of substrate or combination of substrates is suitable for each edible mushroom

species or each strain. Shah ZA et al. (2004) cultivated *P. ostreatus* on mixing or single substrates such as sawdust + wheat straw, sawdust + leaves, wheat straw + leaves, sawdust, wheat straw, leaves and stated that time for completion of spawn running from 16.67 to 23.33 days, the time for formation of fruiting bodies from 27 to 35 days, the largest biological efficiency was on the sawdust substrate (64.69%). However, in the research of Zenebe Girmay et al. (2016), *P. ostreatus* was cultivated on cotton seed, pepper waste, wheat straw and sawdust, with biological efficiency ranging from 9.73 to 74.17%. The lowest biological efficiency was exhibited on sawdust substrate and the highest on the cotton seed substrate (Zenebe Girmay et al., 2016). In the present study, sawdust substrate combined with rice straw or added rice bran and $CaCO_3$ was quite suitable to the growth and development of *P. ostreatus* OM3 (biological efficiency reached from 53.85% to 61.12%).

Table 4. Cultivation *P. ostreatus* OM3 on different substrates

Formulas	Days for mycelium running	Days for forming fruiting bodies	The average mushroom yield (g/bag)	Biological efficiency (%)
FB1	21.10 ± 1.94^{a}	31.97 ± 2.58^{a}	807.7 ± 93.73^{b}	53.85 ± 6.36^{b}
FB2	14.07 ± 1.87^{b}	23.37 ± 2.55^{b}	916.8 ± 76.54^{a}	61.12 ± 5.19^{a}
LSD5%	0.86	1.31	46.84	3.12

Note: Mean values followed by the different letter within a column are statistically different for 0.05 significant level.



Figure 6. The forming fruting bodies of *P. ostreatus* OM3

For the edible mushrooms in Vietnam, the cultivating substrate usually contains a certain amount of sawdust, possibly mixed sawdust, rubber sawdust, or sawdust of *Manglietia conifera*, etc. Besides, rice straw, cotton waste, cob,… are also popular substrates. Truong Nguyen Binh et al. (2019) cultivated grey oyster mushroom on a mixed substrate consisting of rice straw and cotton waste supplemented with 5% rice bran and 0.5% $CaCO_3$. The time for spawn running on substrates was 20.45 days and the highest biological efficiency was 63.94% with 5% inoculation rate. At the inoculation rate of 3%, the spawn running time was 23.24 days and the yield was 59.99%. Compared with this study, the inoculation rate was 2% (30 g spawn/1.5 kg dry substrate), the spawn running time was only 14.07 - 21.10 days, the highest biological efficiency reached 61.12% in the formula containing 48% sawdust + 48% rice straw + 3% rice bran + 1% $CaCO_3$. This biological efficiency is higher than the inoculation rate of 3% and slightly lower than the inoculation rate of 5% of the above research with straw and cotton waste as substrate

4. CONCLUSION

This study presented the results of isolation, spawn propagation (level 1, level 2 and level 3) and experimental cultivation of grey oyster mushroom (*P. ostreatus*). Three strains of *P. ostreatus* were isolated from fruiting bodies. Among them, *P. ostreatus* OM3 (isolated from freshly harvested fruiting bodies) had the best vigor when it was cultured in PGA medium for propagation level 1 (colony diameter: 8.8 cm in 7 days of culture) and PG liquid medium for propagation level 2 (the number of spherical colonies: 860.3 colonies/100 mL; dried biomass: 43.35 g/L). The medium for propagation level 3 forming grain spawn consisting of 99% paddy grain + 1% $CaCO_3$ was the most suitable for *P. ostreatus* OM3. The substrate mixture consisting of 48% sawdust + 48% rice straw + 3% rice bran + 1% $CaCO_3$ powder was also the best for *P. ostreatus* OM3 with the mycelium running time: 14.07 days, fruiting bodies forming time: 23.37 days, mushroom yield: 916.8 g/bag of 2.5 kg substrate, corresponding to 1.5 kg dry substrate; and biological efficiency: 61.12%.

This study provides useful data on isolation, spawn propagation, and cultivation of grey oyster mushroom in particular and edible mushroom in general.

REFERENCES

1. Da Luz J.M.R., Paes, S.A., Nunes, M.D., Torres, D.P., da Silva, M.C.S., 2012. Lignocellulolytic enzyme production of *Pleurotus ostreatus* growth in agroindustrial wastes. *Braz. J. Microbiol.*, 43, 1508-1515.

2. Caglarirmak N., 2007. The nutrients of exotic mushrooms (*Lentinula edodes* and *Pleurotus* species) and an estimated approach to the volatile compounds. *Food Chem.*, 105: 1188-1194.

3. Chorvathova V, Bobek P, Ginter E, Klvanova J., 1993. Effect of the oyster fungus on glycaemia and cholesterolaemia in rats with insulin-dependent diabetes. *Physiological Research*, 42: 175-175.

4. Sivrikaya Hüseyin, Latife Bacak, AkınSaraçbaşı, Ihsan Toroğlu, HüdaverdiEroğlu, 2002. Trace elements in *Pleurotus sajor-caju* cultivated on chemithermomechanical pulp for bio-bleaching. *Food Chemistry*, 79(2): 173-176.

5. R. Mary Josephine, 2015. A review on oyster mushroom (*Pleurotus* spp). *International Journal of Current Research*, Vol. 7, Issue 01: 11225-11227.

6. Mihai R.A., Melo Heras E.J., Florescu L.I., Catana R.D., 2022. The edible gray oyster fungi *Pleurotus ostreatus* (Jacq. ex Fr.) P. Kumm a potent waste consumer, a biofriendly species with antioxidant activity depending on the growth substrate. *J. Fungi* (Basel), 2002 Mar 9, 8(3): 274. https://doi.org/10.3390/jof8030274.

7. Z.A. Shah, M. Ashraf and M. Ishtiaq Ch., 2004. Comparative study on cultivation and yield performance of oyster mushroom (*Pleurotus ostreatus*) on different substrates (wheat straw, leaves, saw dust). *Pakistan Journal of Nutrition,* 3: 158-160.

8. Piškur, B., Bajc, M., Robek, R., Humar, H., Sinjur, I., Kadunc, A., Oven, P.,; Rep, G., Petkovšek, S.A.S., Kraigher, H., et al., 2011. Influence of *Pleurotus ostreatus* inoculation on wood degradation and fungal colonization. *Int. Biodeterior. Biodegrad*, 102: 10611-10617.

9. Girmay Z., Gorems W., Birhanu G. et al*.,* 2016. Growth and yield performance of *Pleurotus ostreatus* (Jacq. Fr.) Kumm (oyster mushroom) on different substrates. *AMB Expr* **6**, 87 (2016). https://doi.org/10.1186/s13568-016-0265-1.

10. Sánchez C., 2010. Cultivation of *Pleurotus ostreatus* and other edible mushrooms. *Appl. Microbiol. Biotechnol*., 85, 1321-1337. [CrossRef] [PubMed]].

11. Truong Binh Nguyen (2004). Rubber tree sawdust - Oyster mushroom cultivation. Mushroom Grower's Handbook.

12. Truong Binh Nguyen, Nguyen Hoang Mai, Phan Hoang Dai, Ngo Thuy Tram, Le Ba Dung, 2019. Oyster mushroom cultivation technique using fermentation substrate. *Journal of Vietnam Agricultural Science and Technology*, 2(99): 93-97.

13. E. Vladimir, 2012. Submerged cultivation of medicinal mushrooms: Bioprocesses and Products*. International Journal of Medicinal Mushrooms*, Vol. 14, No. 3, 211-239.

14. Pham Thi Thu, Le Van Ve, Tran Thu Ha, Nguyen Duy Trinh, 2018. Liquid fermentation technology, its applications in cultivation of golden oyster mushroom (*Pleurotus citrinopileatus*). *Vietnam Journal of Science, Technology and Engineering*, 60(5): 27-33.

15. Le Duy Thang, Tran Van Minh, 2005. Guide to growing mushrooms. *Agricultural Publishing House*, *Vietnam.*

16. Girmay Z., Gorems W., Birhanu G., Zewdie S., 2016. Growth and yield performance of *Pleurotus ostreatus* (Jacq. Fr.) Kumm (oyster mushroom) on different substrates. *AMB Express*, 6: 87.

17. Nguyen Thi Sau, 2009. Mushroom cultivation and processing techniques. *Ho Chi Minh City University of Industrial Technology, Vietnam*.

18. Nguyen Thi Bich Thuy, 2014. Research on some biological properties and appropriate technology for breeding and cultivating some edible and medicinal mushrooms in Vietnam. PhD. thesis. *Vietnam Academy of Agricultural Sciences*.

19. N. Abdullah, R. Ismail, N. Johari, M. Annuar, 2013. Production of liquid spawn of an edible grey oyster mushroom, *Pleurotus pulmonarius* (Fr.) Quél. by submerged fermentation and sporophore yield on rubber wood sawdust. *Scientia Horticulturae*, Vol. 161: 65-69.

20. Ha Thi Quyen, Nghiem Thi Hue, Hoang Thi Hong Nga, Quach Van Son, Le Tien Vuong, Chu Duc Ha, 2022. The effect of static magnetic field on the growth of the milky mushroom (*Calocybe indica*). *Proceeding of Vietnam National Conference on Biotechnology* 2022: 557-563.

21. Tran Thanh Thy and Le Van Vang, 2022. Study appropriate media and subtrates of production Emperor mushroom (*Calocybe indica* APK2). *CTU Journal of Science*, 58(2): 185-193.

22. Nguyen Thi Bich Thuy, Ngo Xuan Nghien, Le Van Ve , Nguyen Thi Luyen, Nguyen Thi Huyen Trang, Phan Thu Huyen, 2019. Mycelial growth and fruting body formation of the king oyster mushroom *Pleurotus eryngii* (DC.: Fr.) strain E2. *Vietnam Journal of Science and Technology*, 61(7): 38-44.

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